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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B. Tech (CBCS) Degree Examinations
May 2025

CC2201-1- ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
(For CC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which of the following is NOT a common task in NLP?
A) Speech recognition B) Machine translation
C) Image classification D) Text summarization
2. Which type of neural network is best suited for processing sequential data, such as language?
A) CNN B) RNN
C) Support Vector Machine (SVM) D) Naïve Bayes Classifier
3. Which sensor is commonly used in autonomous robotic systems for environmental mapping?
A) LiDAR B) GPS
C) Microphone D) Infrared sensor
4. Which of the following is a fundamental component of an expert system?
A) Knowledge base B) Operating system
C) Hardware accelerator D) Compiler
5. Which type of agent program selects actions based solely on the current percept?
A) Model-based agent B) Reflex agent
C) Goal-based agent D) Utility-based agent
6. Which type of agent builds an internal model of the world?
A) Simple reflex agent B) Goal-based agent
C) Utility-based agent D) Model-based reflex agent
7. A well-defined problem in AI consists of which components?
A) Initial state, goal state, and actions B) Sensors, environment, and actions
C) Planning, execution, and evaluation D) Data, algorithm, and implementation
8. Which search method is guaranteed to find the shortest path in an unweighted graph?
A) Depth-first search B) Greedy best-first search
C) Breadth-first search D) Hill climbing
9. What are the data points closest to the decision boundary in SVMs called?
A) Hyperplanes B) Support vectors
C) Centroids D) Decision points
10. What does a hard-margin SVM assume?
A) The data is not linearly separable B) The decision boundary can be nonlinear
C) The margin can be violated to accommodate misclassified points D) The dataset has no overlapping classes
11. What is discretization in machine learning?
A) Converting continuous attributes into categorical bins B) Normalizing features to a common scale
C) Removing irrelevant features D) Increasing dataset complexity

12. Which of the following is a technique for dealing with missing data?
 A) Data augmentation B) One-hot encoding
 C) Imputation D) Dropout
13. The distance metric commonly used in k-NN is:
 A) Cosine similarity B) Euclidean distance
 C) Manhattan distance D) Pearson correlation
14. Bayes' Theorem is used to calculate:
 A) The probability of an event occurring given prior knowledge B) The distance between two probability distributions
 C) The Gini Index in decision trees D) The number of clusters in k-means
15. What assumption does the Naïve Bayes classifier make about features?
 A) They are highly correlated B) They must be normally distributed
 C) They have the same probability distribution D) They are completely independent
16. Which of the following best defines Bayes' Error Rate?
 A) The error rate of Naïve Bayes B) The error rate of k-NN
 C) The error rate of the best possible classifier D) The error rate of the decision tree classifier
17. What does DBSCAN stand for?
 A) Dynamic Bayesian Spatial Clustering Algorithm B) Density-Based Spatial Clustering of Applications with Noise
 C) Data-Based Statistical Clustering of Applications with Noise D) Density-Based Statistical Clustering Algorithm
18. What is the primary goal of the BIRCH algorithm?
 A) To partition data into non-overlapping clusters B) To reduce large datasets into smaller representations for clustering
 C) To detect anomalies in time series data D) To perform real-time classification
19. Which clustering method is best suited for detecting arbitrarily shaped clusters?
 A) DBSCAN B) K-Means
 C) Agglomerative Hierarchical Clustering D) BIRCH
20. Which of the following clustering algorithms is best for handling large datasets efficiently?
 A) K-Means B) Hierarchical Clustering
 C) DBSCAN D) BIRCH

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define in your own words: (i) intelligence, (ii) artificial intelligence, (iii) agent, (iv) rationality, (v) logical reasoning.	8	L*2	1	1,2
	b) Differentiate between Deep Learning and Machine Learning with Suitable examples.	8	L2	1	1,2
2.	a) Outline the PEAS description of the task environment for an automated taxi, Interactive English tutor and Part-picking robot.	8	L2	2	1,2
	b) With Suitable Examples Explain the task environments and their characteristics.	8	L2	2	1,2
3.	a) Discuss how a model-based reflex agent improves upon the limitations of a simple reflex agent.	8	L2	2	1,2
	b) Explain how a problem-solving agent for a vacuum cleaner works. What steps does it take to clean the environment?	8	L2	2	1,2
Unit – II					
4.	a) Explain three types of machine learning techniques. List algorithms for each of the types.	8	L2	3	1,2,3
	b) Explain the decision tree induction for classification with suitable example.	8	L2	3	1,2,3
5.	a) Write an algorithm for k-nearest neighbor algorithm. Explain the characteristics of Nearest- Neighbor Classifiers.	8	L2	3	1,2,3

- b) How does linear SVM work in a separable case? Explain the concept of finding the optimal hyperplane.

8 L2 3 1,2,3

6. a) A database has the following transactions. Assume min_sup = 2. Generate the frequent itemsets by applying Apriori algorithm. List the strongest Association rules for min_cof=75%

TID	Items bought
10	A,B,D
20	A,C,D
30	A,D,E
40	B,E,F
50	B,C,D,E,F

8 L3 4 1,2,3

- b) Apply Hunt's algorithm to generate decision tree for the following dataset.

Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

8 L3 4 1,2,3

Unit – III

7. a) Compare and contrast the performance of K-means clustering and agglomerative hierarchical clustering on the same dataset. When would you choose one over the other?
- b) Analyze the strengths and weaknesses of using DBSCAN for clustering when compared to K-means and hierarchical clustering in terms of handling noise and outliers.
8. a) Examine how the CF tree in the BIRCH algorithm helps improve the scalability of clustering for large datasets. What are the limitations of this approach?
- b) What is cluster analysis, and what are its primary goals in data mining and machine learning?

8 L2 5 1,2,3

8 L2 5 1,2,3

8 L2 5 1,2,3

8 L2 5 1,2,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
